

Project title: **Biodiversity dynamics in coevolutionary meta-ecosystems**

Scientific exchanges: Plan for research stay

The research plan should not exceed 5 pages.

Timeframe of the research stay:	15 th Oct 2018 – 31 st Dec 2018
Host for the visit (researcher and institute):	Carlos J. Melián, EAWAG
Visiting researcher (name and institute):	Cecilia Siliansky de Andreazzi, Fiocruz Mata Atlântica, Fundação Oswaldo Cruz, Rio de Janeiro, Brazil

1. Aims and relevance

One of the main challenges in theoretical and applied ecology is to incorporate the paradigm of meta-ecosystem ecology to understand spatio-temporal ecosystem functioning. Meta-ecosystems are a set of ecosystems connected by spatial flows of energy, materials and organisms across their boundaries (Loreau et al. 2003). The meta-ecosystem concept provides a powerful theoretical tool to understand the emergent properties that arise from spatial coupling of local ecosystems, such as coevolutionary selection mosaics, stabilization of ecosystem processes and indirect interactions at landscape or regional scales. The research visit aims to develop theory and analytical methods to bridge the gap between coevolutionary and ecological spatio-temporal dynamics of species-rich interaction networks. This will advance our comprehension of ecosystem functioning across spatial and temporal scales, focusing on the integration of empirical data with models.

Evolutionary dynamics -the selection pressures affecting traits distributions- and ecological dynamics -the biotic and abiotic factors causing demographic changes on populations- can work out at similar spatio-temporal scales (Hairston et al. 2005, Fussmann et al. 2007, Schoener 2011; Thompson 2013). Rapid evolutionary changes have been invoked as primary forces driving ecosystem function in species-rich meta-ecosystems (Urban et al. 2008, Encinas-Viso et al. 2014, Andreazzi et al. 2017, 2018). Yet, most theories developed to study the connection between evolutionary and ecological dynamics lack explicit reciprocal trait changes among many interacting species –coevolution. This limit us to understand the effects of rapid changes on meta-ecosystems functioning.

Coevolutionary hypotheses and meta-ecosystem theory would improve their predictions on biodiversity dynamics by taking into account the coevolutionary processes occurring at different spatio-temporal scales (Urban and Skelly 2006, Melián et al. 2015a, Frickel et al. 2016). Spatially variable biotic selection and the migration of individuals among heterogeneous patches can generate complex selection mosaics leading to spatial variation in species traits. This leads to spatial variation in the structure and stability of ecological networks (Massol et al. 2017, Gravel et al. 2016, Melián et al., 2018), which in turn may affect the coevolutionary process (Thompson 2005) and the function of meta-ecosystems (Mouquet and Loreau 2003, Leibold et al. 2004, Logue et al. 2011, Winegardner et al. 2012, Borthagaray et al. 2014 a,b, 2015, Astegiano et al. 2015, Melian et al. 2015b, Laroche et al.

2016, Massol et al. 2017 a, b). Metacommunity and meta-ecosystem theory help us to understand how spatially integrated processes mainly governed by dispersal (mass effect, neutral processes) or niche width (species sorting, patch dynamics) can alter biodiversity spatial and temporal patterns. Incorporating the meta-ecosystem perspective into existing models of coevolutionary interaction networks (Guimarães et al. 2011, Andreazzi et al. 2017) will improve our ability to predict biodiversity distribution across space and time and ecosystem responses to global changes.

Most interactions between species generate reciprocal selection pressures. The coevolutionary dynamics that emerges over time varies with the fitness consequences of the ecological interaction (competitive, mutualistic, or antagonistic) and the architecture of the coevolving interactions (Wade 2007, Loeuille and Leibold 2008, Doebeli 2011, Nuismer et al. 2013, Althoff et al. 2014, Hembry et al. 2014, Althoff and Drossel 2016). There are empirical and theoretical evidence showing that ecological interactions drives coevolutionary dynamics in local communities (Guimarães et al. 2011, Andreazzi et al. 2017, Koskella & Brockhurst 2014, Hembry et al. 2014). However, the connection between traits and coevolution at small scales and the large-scale associations with biodiversity function and meta-ecosystem dynamics are not well understood (Wade 2007, Doebeli 2011, Althoff et al. 2014, Hembry et al., 2014). Thus, developing theory and analytical methods that merge coevolutionary dynamics among many interacting species at different spatial scales is critical to our understanding of the web of life.

During the research visit, we aim to integrate coevolutionary processes happening in multiple ecological interactions into meta-ecosystem theory. We will study the interdependent effects of the coevolution of traits, biodiversity dynamics and ecosystem function in an interaction network context. Making the connection between traits and demographic dynamics will enable us to assess how the speed of reciprocal trait change drives the dynamics of species interaction networks and ecosystem function. Meta-ecosystem dynamics and coevolutionary processes can be integrated by explicitly linking species traits to the fitness effects of the interactions (Nuismer et al. 2013, Georgelin and Loeuille 2014). This connection is challenging because there is no existing method designed to take into account the fitness consequences of many interacting individuals and species in meta-ecosystems (De Long and Gibert 2016).

2. Methods

We propose to extend the meta-ecosystem theory to study how reciprocal trait changes among many interacting species might affect biodiversity dynamics and ecosystem function. This overarching goal will be achieved by focusing on two main research questions:

Q1: *How does reciprocal trait change influence biodiversity dynamics, species extinctions and ecosystem function in species-rich meta-ecosystems?*

Q2: *How do the spatial arrangement of ecosystems and coevolutionary dynamics interact to facilitate rapid evolutionary changes in meta-ecosystems?*

During the research visit period we will develop the coevolutionary meta-ecosystem model to study feedbacks of reciprocal trait change in spatially explicit heterogeneous environments. Our model will include four interacting hierarchical levels (individual, population, ecosystem, and meta-ecosystem) (Melian et al., 2018). The basic model will represent a meta-ecosystem with a network of P patches and S species per patch each

containing initially N_i individuals per species i (Gravel et al. 2016). Each species i in patch P undergoes trait dynamics driven by mating, gene flow (i.e., migration between patches), recombination, interaction selection (i.e., which drives the speed of reciprocal trait change) and environmental selection following an optimum for each species i . The distribution of traits of each species i will contain many phenotypes, each associated to a changing fitness value. We will track species trait coevolution and changes associated to ecosystem function using abundances and traits that drive the stability in each local patch. We will explicitly consider trait variance to calculate the fitness of a given trait in species-rich systems using the eco-evolutionary Gillespie algorithm (De Long and Gibert 2016). Using this model we will explore: 1) How different strengths of coevolutionary selection (i.e., reciprocal trait change, **Q1**), influence biodiversity dynamics in each local community and in the meta-ecosystem, 2) How different scenarios of reciprocal trait change (i.e., trait matching, trait mismatching, and exploitation barriers) modulate the number of coexisting species (**Q1**), and 3) How different spatial arrangements of interacting communities might facilitate or inhibit rapid changes in biodiversity and ecosystem function (**Q2**).

We will organize a workshop (see the Biodiversity Dynamics in Coevolutionary Meta-ecosystems workshop proposal) to discuss the development of new theoretical and analytical approaches in metacommunity and meta-ecosystem theories and how to join them to coevolutionary dynamics among multiple interacting species. We will also discuss how to apply the coevolutionary meta-ecosystem approach to current problems in epidemiology, agroecology and conservation biology.

3. Schedule

The main activities of the research visit are focused in the following topics. A detailed cronogram of the activities is shown in Table 1.

- (1) Organize and participate in the coevolutionary meta-ecosystems workshop: During the workshop we will discuss future directions that unify coevolutionary and meta-ecosystem theories and methods for integrating theory with empirical data;
- (2) Development of the mathematical model: We will develop mathematical approaches that incorporate the meta-ecosystem perspective into existing models of coevolving interaction networks;
- (3) Analysis of the mathematical model: Calculate and evaluate the analytical solutions for the model.
- (4) Implement the mathematical model: Develop the model algorithm and code.
- (5) Design and implement the simulations: Analyse model parameter space, define relevant ecological and evolutive scenarios, run the simulations.
- (6) Link empirical data with model: Use empirical spatio-temporal trait data to evaluate model predictions.
- (7) Draft manuscript: Start drafting a working paper about Biodiversity dynamics and ecosystem functioning in coevolutionary meta-ecosystems emphasizing its synthesis aims between theory and data and fundamental and applied research.

Table 1: Cronogram of the research visitXactivity/ weeks			
	October	November	December
1			
2			
3			
4			
5			
6			
7			

4. Expected outcomes

This collaboration will produce high-impact results by unifying ecological and evolutionary processes across levels of biological organization, such as networks of species interactions and meta-ecosystems. Our approach will focus on:

- (1) Demonstrate how the speed of reciprocal trait changes at local scales may be responsible for large-scale associations between coevolution and biodiversity dynamics and therefore meta-ecosystem functioning;
- (2) Improvement of our predictive understanding of the mechanisms driving reciprocal trait change, interaction strength between species and biodiversity dynamics, which affects the stability of meta-ecosystems;
- (3) Use of empirical data for testing model predictions for the metaecosystems responses to global changes and evaluating the advantages of using this approach to current problems in epidemiology, agroecology and conservation biology.
- (4) Strengthen our knowledge about the interactions between micro-coevolutionary processes and macro-evolutionary patterns of biodiversity, across spatio-temporal scales.

5. Partnership: added value for visiting fellow and host institute and potential for further collaboration

The research visit will enable the theoretical and analytical development and test of the coevolutionary meta-ecosystem model. The visiting fellow is a researcher with background in wildlife disease ecology, ecology and evolution of species interactions and network theory. She has experience in participating in multidisciplinary collaborative projects and her expertise complements the research developed by the host research group. The collaborative work with Carlos Melián and all the researchers participating in the coevolutionary meta-ecosystems workshop will improve her expertise in metacommunity and meta-ecosystem theory, and in the analytical methods applied to evolutionary ecology.

By promoting this collaboration, the host institution will assume a leadership role in developing theory that integrates coevolutionary dynamics and meta-ecosystem functioning with an explicit applied perspective. The host and visiting researches have the ideal background to complement each other and this collaboration will have a decisive impact in the visiting resercher career and in the development of the field. The added value of the

workshop will be guaranteed by merging researchers with strong skills in field studies and empirical patterns, evolutionary ecology, meta-ecosystems, network theory and computational and analytical tools.

6. Experience of the applicants

Carlos Melián is a researcher from EAWAG since 2010. He is a theoretical evolutionary ecologist working on the integration of methods accounting for different biological organization levels through stochastic models of biodiversity. Dr Melián has been a postdoc at the National Center for Ecological Analysis and Synthesis, University of California Santa Barbara. He is authored dozens of peer-reviewed publications including high-impact journals such as PNAS and Trends in Ecology and Evolution. The host researcher supervised postdocs and PhD students and his research programme follows two main axes: the mechanisms connecting trait dynamics to biodiversity and food webs, and the mechanisms promoting radiations and diversification patterns. Dr. Melián is currently head of the Fish Ecology and Evolution department at Eawag, Kastanienbaum.

Cecilia S. Andreazzi is a researcher at the Fundação Oswaldo Cruz. The visiting fellow applicant is an empirical and theoretical ecologist with background in wildlife disease ecology, evolution and ecology of species interactions and network theory. During her PhD she developed mathematical models to study evolution and coevolution in species-rich interaction networks. She is now leading a project aiming to understand how the functional diversity of wild mammal hosts is related to the functional diversity of their parasites and the network of host-parasite interactions at local and regional scales. The aim of this project is to predict zoonotic infectious disease risk in metaecosystems. The applicant is interested in understanding the evolution of species-rich ecological networks and the effects of environmental changes on host-parasite interactions and infectious disease dynamics. She has experience with multidisciplinary collaboration including research in complex systems.

7. References

- Allhoff, K.T. and Drossel, B. (2013). When do evolutionary food web models generate complex networks? *J. Theor. Biol.*, 334:122–129.
- Althoff, D. M., Segraves, K. A., and Johnson, M. T. J. (2014). Testing for coevolutionary diversification: linking pattern with process. *Trends in Ecology and Evolution*, 29:82-89.
- Andreazzi, C. S., Thompson, J. N. and Guimarães, P. R. (2017). **Network structure and selection asymmetry drive coevolution in species-rich antagonistic interactions**, *The American Naturalist*, 190(1): 99-115.
- Andreazzi, C. S., Guimarães, P. R., and Melián, C. J. (2018). **Eco-evolutionary feedbacks promote fluctuating selection and long-term stability of antagonistic networks**, *Proceedings of the Royal Society B: Biological science*, 285:20172596.
- Astegiano, J., Guimarães Jr, P. R., Cheptou, P. O., Vidal, M. M., Mandai, C. Y., Ashworth, L., and Massol, F. (2015). Persistence of plants and pollinators in the face of habitat loss: insights from trait-based metacommunity models. *Advances in Ecological Research*, 53:201-257.
- Borthagaray, A. I., Barreneche, J. M., Abades, S., and Arim, M. (2014a) Modularity along organism dispersal gradients challenges a prevailing view of abrupt transitions in animal landscape perception. *Ecography*, 37:564-571.
- Borthagaray, A. I., Arim, M., and Marquet, P. (2014b). Inferring species roles in metacommunity structure from species coexistence networks. *Proceedings of the Royal Society B*, 281:20141425.
- Borthagaray, A. I., Pinelli, V., Berazategui, M., Rodriguez-Tricot, L., & Arim, M. (2015) Effects of metacommunity networks on local community structures: from theoretical predictions to empirical evaluations. *Aquatic functional biodiversity: an ecological and evolutionary perspective*. Academic Press, Cambridge, 75-111.

- DeLong, J.P. and Gibert, J.P. (2016) Gillespie eco-evolutionary models (GEMs) reveal the role of heritable trait variation in eco-evolutionary dynamics. *Ecol. Evol.*, 6:935–945.
- Doebeli, M. (2011). Adaptive Diversification. Princeton Univ. Press, Princeton.
- Encinas-Viso, F., Revilla, T. A., and Etienne, R. S. (2014). Shifts in pollinator population structure may jeopardize pollination service. *Journal of Theoretical Biology*, 352, 24-30.
- Frickel, J., Sieber, M., and Becks, L. (2016). Eco-evolutionary dynamics in a coevolving host–virus system. *Ecology letters*, 19:450-459.
- Fussmann, G. F., Loreau, M. and Abrams, P. A. (2007), Eco-evolutionary dynamics of communities and ecosystems. *Functional Ecology*, 21:465-477.
- Georgelin, E., and Loeuille, N. (2014). Dynamics of coupled mutualistic and antagonistic interactions, and their implications for ecosystem management. *Journal of theoretical biology*, 346, 67-74.
- Gravel, D., Massol, F., and Leibold, M. A. (2016). Stability and complexity in model meta-ecosystems. *Nature communications*, 7, 12457.
- Guimarães, P.R., Jordano, P. and Thompson, J.N. (2011). Evolution and coevolution in mutualistic networks. *Ecology Letters*, 14:877–85.
- H Hairston, N.G., Ellner, S.P., Geber, M.A., Yoshida, T. and Fox, J.A. (2005) Rapid evolution and the convergence of ecological and evolutionary time. *Ecology Letters*, 8:1114–1127.
- Hembry, D. H., Yoder, J. B., and Goodman, K. R. (2014). Coevolution and the diversification of life. *The American Naturalist*, 184:425-438.
- Koskella, B., & Brockhurst, M. A. (2014). Bacteria–phage coevolution as a driver of ecological and evolutionary processes in microbial communities. *FEMS microbiology reviews*, 38(5), 916-931.
- Laroche, F., Jarne, P., Perrot, T., & Massol, F. (2016) The evolution of the competition-dispersal trade-off affects α and β -diversity in a heterogeneous metacommunity. *Proceedings of the Royal Society Biological Sciences Series B*, 283:20160548.
- Leibold, M.A., Holyoak, M., Mouquet, N., Amarasekare, P., Chase, J. M., Hoopes, M. F., Holt, R. D., Shurin, J. B., Law, R., Tilman, D., Loreau, M., & Gonzalez, A. (2004) The metacommunity concept: a framework for multi-scale community ecology. *Ecology Letters*, 7:601–613.
- Loeuille, N., & Leibold, M. A. (2008). Evolution in metacommunities: on the relative importance of species sorting and monopolization in structuring communities. *The American Naturalist*, 171(6), 788-799.
- Logue, J. B., Mouquet, N., Peter, H., Hillebrand, H., & Metacommunity Working Group. (2011) Empirical approaches to metacommunities: a review and comparison with theory. *Trends in ecology & evolution*, 26:482-491.
- Loreau, M., Mouquet, N., & Holt, R. D. (2003). Metaecosystems: a theoretical framework for a spatial ecosystem ecology. *Ecology Letters*, 6:673-679.
- Massol, F., Altermatt, F., Gounand, I., Gravel, D., Leibold, M., & Mouquet, N. (2017a) How life-history traits affect ecosystem properties: effects of dispersal in metaecosystems. *Oikos*, 126:532-546.
- Massol, F., Dubart, M., Calcagno, V., Cazelles, K., Jacquet, C., Kéfi, S., & Gravel, D. (2017b) Island biogeography of food webs. *Advances in Ecological Research*, 56:183-262.
- Melián, C. J., Matthews, B., Andreazzi, C. S., Rodríguez, J. P., Harmon, L. H., and Fortuna, M. A. (2018). Deciphering the interdependence between ecological and evolutionary networks. *Trends in Ecology and Evolution* (In Press).
- Melián, C. J., Seehausen, O., Eguíluz, V. M., Fortuna, M. A., & Deiner, K. (2015a). Diversification and biodiversity dynamics of hot and cold spots. *Ecography*, 38(4), 393-401.
- Melián, C. J., Krivan, V., Altermatt, F., Starý, P., Pellissier, L., & De Laender, F. (2015). Dispersal dynamics in food webs. *The American Naturalist*, 185(2), 157-168.
- Mouquet, N. & Loreau, M. (2003). Community patterns in source-sink metacommunities. *The American Naturalist*, 162:544–557.
- Nuismer, S., Jordano, P. and Bascompte, J. (2013). Coevolution and the architecture of mutualistic networks. *Evolution*, 67:338–354.
- Thompson, J. N. (2005). The geographic mosaic of coevolution. University of Chicago Press. Chicago.
- Thompson, J. N. (2013). Relentless Evolution. University of Chicago Press. Chicago.
- Urban, M. C., & Skelly, D. K. (2006) Evolving metacommunities: toward an evolutionary perspective on metacommunities. *Ecology*, 87:1616-1626.
- Urban, M.C., Leibold, M.A., Amarasekare, P., De Meester, L., Gomulkiewicz, R., Hochberg, M.E., Klausmeier, C.A., Loeuille, N., De Mazancourt, C., Norberg, J. and Pantel, J.H., (2008). The evolutionary ecology of metacommunities. *Trends in ecology & evolution*, 23(6): 311-317.
- Wade, M. J. (2007). The co-evolutionary genetics of ecological communities. *Nature reviews*, 8:185-195.
- Winegardner, A. K., Jones, B. K., Ng, I. S., Siqueira, T., & Cottenie, K. (2012). The terminology of

metacommunity ecology. *Trends in ecology & evolution*, 27:253-254.